SUB-GAUSSIAN CONCENTRATION FROM RANDOM YOUNG DIAGRAMS

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Abstract. Consider a uniform random partition of n. We draw its Young diagram, place the diagram so that the corner is at $(0,0)$ in coordinate plane, and scale the diagram so that each square has size $\frac{1}{\sqrt{n}} \times \frac{1}{\sqrt{n}}$. We prove that (i) the height of the scaled diagram at $x = t$ for any t satisfies a sub-Gaussian concentration with standard deviation $O(n^{-1/6+\varepsilon})$, and (ii) the expected height converges to $t - \frac{\sqrt{6}}{\pi}\log(e^{\frac{\pi}{\sqrt{6}}t} - 1)$ as $n \to \infty$. Our results explain why the scaled Young diagram converges to the area under the curve $(e^{\frac{\pi}{\sqrt{6}}x} - 1)(e^{\frac{\pi}{\sqrt{6}}y} - 1) = 1$ with high probability.

1. Introduction

Let n be a nonnegative integer. A **partition** of n is a sequence $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k > 0$ of positive integers such that $\lambda_1 + \lambda_2 + \cdots + \lambda_k = n$. For example, there are 7 partitions of $n = 5$, namely

$$5, \quad 4+1, \quad 3+2, \quad 3+1+1, \quad 2+2+1, \quad 2+1+1+1, \quad \text{and} \quad 1+1+1+1+1.$$

Observe that partitions can also be seen as a *unordered* positive integers that sum to n. We say that the number of *parts* of a partition is the number of integers we are adding together, and a *part* is any one of the integers. For example, the partition $6+3+3+2+1$ of 5 has 5 parts, each of sizes 6, 3, 3, 2, and 1.

A powerful tool for studying partitions and their properties is the Young diagram.

Definition 1. A **Young diagram** is a finite collection of 1×1 squares left-aligned and bottom-aligned such that the i-th row from the bottom has λ_i many boxes. In total, the number of non-empty rows will be k, the number of parts. For our purposes, we will pick the axis, so that the bottom row and the left column just touch the x- and y-axis respectively.

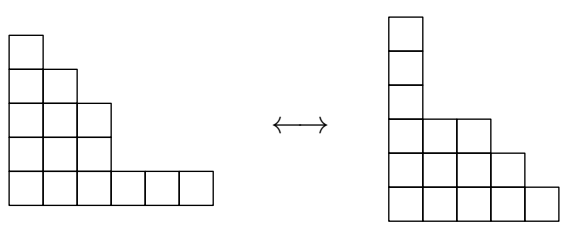

Figure 1. Left: the Young diagram of partition $6+3+2+2+1$ of 15. Right: reflecting across the diagonal gives the Young diagram of another partition $5+4+3+1+1+1$ of 15.

The Young diagram is a useful way to visualize partition. Most immediately, it creates a symmetry in the space of partitions as shown in Figure 1 that reflecting a Young diagram

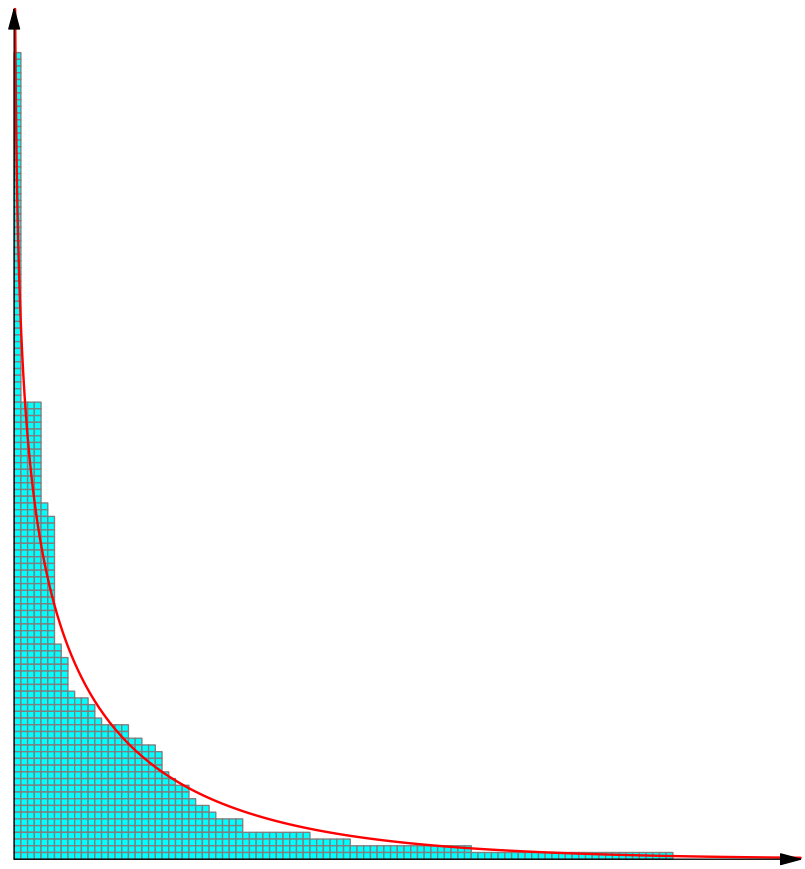

FIGURE 2. A scaled Young diagram of a random partition of 1000, with the limiting curve in red.

across the diagonal gives Young diagram of another partition. Young diagrams are also extensively used in the study of representations of symmetric groups.

This paper will study what the structure of an "random" partition is by considering the Young diagram it generates. That is, we will take a uniformly randomly chosen partition of n and consider its Young diagram. We study the region that the squares in the Young diagram cover.

The area of a Young diagram of a partition of n is, of course, n. So to properly study a "random" partition, we will scale both axis by $1/\sqrt{n}$ to insures that the area in the resulting diagram is always constantly 1.

Definition 2. A **scaled Young diagram** is a young diagram, where the axis are scaled by $1/\sqrt{n}$, while the bottom row and the left column still touch the x- and y-axis respectively.

Our goal is to show with high probability, the region that a scaled Young diagram covers converges to a region bounded below by some smooth curve. To state what we meant by that, we introduce a random variable $\boldsymbol{H_t}$ to be the height of the scaled Young diagram at x-coordinate t. Observe that H_t is equal to $\frac{1}{\sqrt{n}}$ times the number of parts of size at least $t\sqrt{n}$.

To figure out what the curve looks like, we compute the expected value of H_t. Our first result gives the formula for that.

Theorem 3. *For a uniform random partition, the expected value of the height of the scaled Young diagram at t, H_t, is given by*

$$\mathbb{E}[H_t] \;=\; \left(t - \frac{\sqrt{6}}{\pi}\log\left(e^{\frac{\pi}{\sqrt{6}}t} - 1\right)\right) + O\left(\frac{\log n}{\sqrt{n}}\right)$$

where the hidden constant in big-O can depend on t.

This expected value calculation suggests that the region that a scaled Young diagram covers converges to the area under the curve

$$y = x - \frac{\sqrt{6}}{\pi}\log\left(e^{\frac{\pi}{\sqrt{6}}x} - 1\right),$$

as shown in Figure 2. This curve can be written in a manifestly symmetric form as

$$\left(e^{\frac{\pi}{\sqrt{6}}x} - 1\right)\left(e^{\frac{\pi}{\sqrt{6}}y} - 1\right) = 1.$$

We note (without proof) that the area under this curve is exactly 1. This can be shown by integrating the curve equation and using the series expansion of logarithm.

Our next result shows that the random variable H_t has a sub-Gaussian distribution with standard deviation $O(n^{-1/6+\varepsilon})$ for any $\varepsilon > 0$.

Theorem 4. *For any $\varepsilon > 0$, there exists some constant $C = C_{t,\varepsilon}$ such that*

$$\mathbb{P}\left[\left|H_t - \mathbb{E}[H_t]\right| \geq \lambda\right] \;<\; \exp\left(\frac{-\lambda^2}{Cn^{-1/3+\varepsilon}}\right)$$

for all $\lambda > \frac{1}{n^6}$. (Note that the exponent 6 in $\frac{1}{n^6}$ can be changed to any positive number.)

A typical approach in showing concentration result like the one above is to break apart the target random variable (H_t in our case) in a sum of independent random variables, and then apply variants of Chernoff's inequality. In our case, a natural way to break apart H_t is to write $H_t = \frac{1}{\sqrt{n}}\sum_{i\geq t\sqrt{n}} X_i$, where X_i is the number of part of size exactly i in the partition. Unfortunately, the X_i's are not independent (they are only *mostly* independent), so we have to use stronger results from probability theory, namely the Azuma's inequality for martingales (Theorem 12).

Still, Azuma's inequality alone is not enough to prove Theorem 4 directly. We discuss the difficulty in detail in Section 4. To very briefly summarize, the issues arising include (but are not limited to) when X_i is revealed to be very large, screwing up the conditional expected value and violating the main condition in Azuma's inequality. To overcome this difficulty, we have to show that those bad events have low probability, condition away from those, and verify that our new martingale satisfies the condition to apply Azuma's inequality.

The standard deviation bound of $O(n^{-1/6+\varepsilon})$ solely comes from the same error term in the asymptotic formula for partitions in Theorem 7. That is, error terms from other sources are sufficiently small so that if the error term $O(n^{-1/6+\varepsilon})$ in Theorem 7 is changed to something sufficiently small (say $O(n^{-1/2})$), then we would get the standard deviation bound of $O(\operatorname{polylog} n/\sqrt{n})$ instead, which is likely much closer to the reality. Given that our proof relies so much on that asymptotic formula, we believe that our standard deviation bound cannot be improved easily.

1.1. **Organization of this paper.** In Section 2 we introduce some notation and the theorems that will allow us to prove concentration. In Section 3 we prove Theorem 3, which computes the expected value of the height of the curve.

The rest of the paper proves Theorem 4. In Section 4, we outline the proof, introduce many of the necessary variables, define the bad events, and completing the proof modulo two propositions: Proposition 15 and Proposition 18. Section 5 proves Proposition 15, which says that we can avoid bad events with high probability. Finally, Section 6 proves Proposition 18, which justify the condition necessary to use Azuma's inequality.

2. Preliminaries

2.1. **Notation.** We fix some notations that will be used throughout the paper.

Partition numbers. Let $p(n)$ be the number of partitions of n and $p(n,k)$ be the number of partitions of n into at most k parts. We extend the definition so that

- $p(n) = p(n,k) = 0$ for all $n < 0$.
- $p(n) = p(\lfloor n \rfloor)$ and $p(n,k) = p(\lfloor n \rfloor, \lfloor k \rfloor)$ when n or k are not integers.

We note the following property of $p(n,k)$.

Proposition 5. *The number of partitions of n whose every part is at most k is also $p(n,k)$.*

Proof. Note that there is a bijection between

$$\begin{aligned}\{\text{partitions of } n \text{ into at most } k \text{ parts}\} &\longleftrightarrow \{\text{Young diagram with at most } k \text{ rows}\}\\ &\longleftrightarrow \{\text{Young diagram with at most } k \text{ columns}\}\\ &\longleftrightarrow \{\text{partitions of } n \text{ whose every part is at most } k\}.\end{aligned}$$

The first and third bijection is given by the transformation from partitions to Young diagram given in Definition 1. In particular, the number of parts in a partition is equal to the number of rows in the Young diagram, and the largest part in a partition is equal to the number of columns in the Young diagram. The second bijection is given by reflecting the Young diagram across the diagonal as shown in Figure 1. □

Summation. We will use the notation

$$\sum_{i=a}^{b} s(i) \quad \text{to denote} \quad \sum_{i=\lceil a \rceil}^{\lfloor b \rfloor} s(i)$$

when a and b are not integers.

Constants and asymptotics notations. We use standard asymptotics notation, most prominently the big O-notation. In particular, we write $f(n) = O(g(n))$ if there exists a constant C and M such that $|f(n)| \leq C|g(n)|$ for all $n > M$. We sometimes write $f(n) \ll g(n)$ for $f(n) = O(g(n))$ as well.

Unless otherwise specified, the hidden constants in big-O notation are allowed to depend on t but no other variables (such as n, k, i, q, etc.) Similarly, unless otherwise specified, all constants (denoted by C, c, D, d, K, ε, etc.) are allowed to depend on t but no other variables. Some letters might be reused. We let ε denote an arbitrary positive number and allow its value to change from line to line.

All variables are positive unless otherwise specified.

2.2. **Asymptotics formula of partition numbers.** The asymptotic formula for $p(n)$ and $p(n,k)$ are well-studied. The first asymptotic formula for $p(n)$ was given by by Hardy and Ramanujan [HR00]. The proof was later simplified by Newman [New62].

Theorem 6. *We have*

$$p(n) \;=\; \frac{1}{4\sqrt{3}\,n}\exp\left(\pi\sqrt{\frac{2n}{3}}\right)\left(1+O\left(\frac{1}{\sqrt{n}}\right)\right).$$

The asymptotics formula for $p(n,k)$ is more complicated. The first proof is given by Szekeres in two separate papers [Sze51; Sze53] that prove the formula for complementary ranges of k. Then Canfield [Can97] gave an elementary proof. The result is as follows.

Theorem 7 ([Can97])**.** *Fix any $\varepsilon > 0$. Then uniformly for all $k \geq n^{1/6}$, we have*

$$p(n,k) \;=\; \frac{f(u)}{n}\cdot\exp\left(n^{1/2}g(u)\right)\cdot(1+O(n^{-1/6+\varepsilon})),$$

where $u = k/n^{1/2}$, and the functions $f(u)$ and $g(u)$ are defined by

$$\begin{aligned} f(u) \;&=\; \frac{v}{2^{3/2}\pi u}\left(1-e^{-v}-\tfrac{1}{2}u^2e^{-v}\right)^{-1/2} \\ g(u) \;&=\; \frac{2v}{u}-u\log(1-e^{-v}), \end{aligned}$$

and $v = v(u)$ is defined implicitly by

$$u^2 \;=\; \frac{v^2}{\int_0^v \frac{x}{e^x-1}\,dx}.$$

(The right hand side is monotone in v, so there exists a unique solution $v > 0$.)

The functions f and g are complicated, but the only important properties that we need is some notion of continuity. It turns out that for f, Lipschitz continuity is the correct condition that we need here.

Definition 8. Let U be a subset of $\mathbb{R}$ and $C > 0$ be a constant. A function $f\colon U \to \mathbb{R}$ is C-**Lipschitz** if and only if

$$|f(x)-f(y)| < C|x-y|$$

for all $x, y \in U$.

The necessary properties of f and g are summarized in the following proposition.

Proposition 9. *The following properties of f and g hold.*

(a) f is bounded, and there exist a constant C such that f is C-Lipschitz.
(b) The function g is strictly increasing and $\lim_{u\to\infty} g(u) = \pi\sqrt{2/3}$.
(c) Put $h(x) = x\,g(1/x)$. Then
- *h is strictly increasing*
- *h has first and second derivatives that are both continuous and bounded.*

Proof. This is a long, tedious, and unenlightening computation. Deferred to Section A. □

2.3. **Martingales and Azuma's inequality.** In this subsection, we review some background about martingale and martingale concentration inequality. The following informal definition suffices for our purpose.

Definition 10. A sequence of random real variables $Z_0, Z_1, Z_2, \ldots, Z_n$ is a **martingale** if for all i,

$$\mathbb{E}[Z_{i+1} \mid Z_0, Z_1, \ldots, Z_i] \;=\; Z_i.$$

A key example of martingale that we will use is the following.

Example 11. Given any sequence of random real variable $X_1, \ldots, X_n$ and any function $f(X_1, \ldots, X_n)$, the **Doob martingale** is defined by

$$Z_i \;=\; \mathbb{E}[f(X_1, \ldots, X_n) \mid X_1, X_2, \ldots, X_{i-1}].$$

In particular, note that $Z_0 = \mathbb{E}[f(X_1, \ldots, X_n)]$.

A martingale allows us to derive inequalities from non-independent random variables that decay exponentially.

Theorem 12 (Azuma's Inequality)**.** *Let $Z_0, Z_1, \ldots, Z_n$ be a martingale and $c > 0$ be a real number such that*

$$|Z_i - Z_{i-1}| \;<\; c, \tag{1}$$

holds for all i. Then for all $\lambda > 0$, we have

$$\mathbb{P}[|Z_n - Z_0| \geq \lambda] \;\leq\; \exp\left(-\frac{\lambda^2}{2nc^2}\right).$$

Proof. See [AS00, Thm. 7.2.1] or many other probability textbooks. □

3. Calculation of Expected Height

This section will show the expected value of H_t converges as predicted in Theorem 3. To set this up, fix a real number $t > 0$, take a uniformly random partition $\lambda = (\lambda_1, \ldots, \lambda_r)$ of n, and let X_i be the number of i's in that partition (i.e., the number of i's that appear among $\lambda_1, \ldots, \lambda_r$). Note that the height of the Young diagram at t is

$$H_t = \frac{1}{\sqrt{n}} \sum_{i=t\sqrt{n}}^{n} X_i.$$

Thus, to compute the expected value $\mathbb{E}[H_t]$, it suffices to compute the expected value $\mathbb{E}[X_i]$ individually and sum them.

We begin by giving a closed form of $\mathbb{E}[X_i]$.

Lemma 13. *We have*

$$\mathbb{E}[X_i] \;=\; \frac{p(n-i) + p(n-2i) + p(n-3i) + \cdots}{p(n)}.$$

We present two proofs.

Algebraic Proof. Note that the sum of the numbers of i across all partitions is the coefficient of x^n in the expression

$$\frac{1}{1-x} \cdot \frac{1}{1-x^2} \cdots \frac{1}{1-x^{k-1}} \cdot (x^k + 2x^{2k} + 3x^{3k} + \ldots) \cdot \frac{1}{1-x^{k+1}} \cdots.$$

Since $x^i + 2x^{2i} + 3x^{3i} + \cdots = \frac{x^k}{(1-x^k)^2}$, we get that this expression is equal to

$$\frac{x^i}{1-x^i}\prod_{m=1}^{\infty}\frac{1}{1-x^m} \;=\; (x^i + x^{2i} + \dots)\left(\sum_{m=1}^{\infty} p(m)x^m\right),$$

so the coefficient of x^n is

$$p(n-i) + p(n-2i) + p(n-3i) + \dots. \qquad \square$$

Combinatorial Proof. Let A_k be the set of partitions of n that have exactly k parts of size i and let B_k be the set of partitions of n that have $\geq k$ parts of size i. We have $B_k \setminus B_{k-1} = A_k$ and $B_{k-i} \subseteq B_k$. Thus,

$$\mathbb{E}[X_i] \;=\; \frac{1}{p(n)}\sum_{k=0}^{\infty} i\#A_k \;=\; \frac{1}{p(n)}\sum_{i=k}^{\infty} k(\#B_k - \#B_{k-1}) \;=\; \frac{1}{p(n)}\sum_{k=1}^{\infty} B_k. \tag{2}$$

Now, a partition of n with at least k parts of size i can be bijected with partitions of $n-ki$ simply by adding or removing exactly k parts of size i. Thus $\#B_i = p(n-ki)$. $\square$

We now estimate $\mathbb{E}[X_k]$. In this lemma, all estimates are uniform in t. In particular, the hidden constant in the big-O notation does not depend on t.

Lemma 14. *Suppose that $i = t\sqrt{n}$ for some constant t. Then we have*

$$\mathbb{E}[X_i] \;=\; \frac{1}{\exp\left(\frac{\pi}{\sqrt{6}}t\right) - 1}\left(1 + O\left(\frac{\log n}{\sqrt{n}}\right)\right).$$

Proof. From the expected value formula (Lemma 13), we get that

$$\mathbb{E}[X_i] \;=\; \frac{1}{p(n)}\sum_{j=1}^{\lfloor n/k\rfloor} p(n-ji)$$

We split this sum into $ji > 10\log n\sqrt{n}$ or not, giving us S_1 and S_2, with $\mathbb{E}[X_i] = S_1 + S_2$, where each S_ℓ is a sum of non-negative terms

$$\begin{aligned}
S_1 \;&=\; \frac{1}{p(n)}\sum_{j=1}^{10\log n\sqrt{n}} p(n-ji) \\
&=\; \sum_{j=1}^{10\log n\sqrt{n}} \frac{n}{n-ji}\exp\left(\pi\sqrt{\tfrac{2}{3}}\left(\sqrt{n-ji}-\sqrt{n}\right)\right)\left(1 + O\left(\frac{1}{\sqrt{n-ji}}\right)\right) \\
&=\; \sum_{j=1}^{10\log n\sqrt{n}} \exp\left(\pi\sqrt{\tfrac{2}{3}}\left(\sqrt{n-ji}-\sqrt{n}\right)\right)\left(1 + O\left(\frac{\log n}{\sqrt{n}}\right)\right) \\
S_2 \;&=\; \frac{1}{p(n)}\sum_{j=10\log n\sqrt{n}}^{n} p(n-ji),
\end{aligned}$$

where we used Ramanujan's formula (Theorem 6) in estimating S_1. Now, we upper bound S_2 by the following

$$S_2 \;\leq\; O\left(n\cdot\frac{p(n-10\log n\sqrt{n})}{p(n)}\right) \;\leq\; O(n\exp(-5\log n)) \;=\; O(n^{-2}).$$

Therefore,

$$\mathbb{E}[X_i] \;=\; O(n^{-2}) + \sum_{j=1}^{10\log n\sqrt{n}} \exp\left(\pi\sqrt{\frac{2}{3}}\left(\sqrt{n - jt\sqrt{n}} - \sqrt{n}\right)\right)\left(1 + O\left(\frac{\log n}{\sqrt{n}}\right)\right).$$

Note that

$$\begin{aligned}\sqrt{n - jt\sqrt{n}} \;&=\; \sqrt{n}\left(\sqrt{1 - \frac{jt}{\sqrt{n}}}\right) \;=\; \sqrt{n}\left(1 - \frac{jt}{2\sqrt{n}}\right)\left(1 + O\left(\frac{j^2t^2}{n}\right)\right)\\ &=\; \left(\sqrt{n} - \frac{jt}{2}\right)\left(1 + O\left(\frac{j^2t^2}{n}\right)\right)\end{aligned}$$

by taking the approximation $\sqrt{1-x} \approx 1 - \frac{x}{2}$ provided by the linear Taylor polynomial and bounding the error by the subsequent derivative. The sum now simplifies to

$$\begin{aligned}\mathbb{E}[X_i] \;&=\; O(n^{-2}) + \sum_{j=1}^{10\log n\sqrt{n}} \exp\left(-\pi\sqrt{\frac{2}{3}}\cdot\frac{jt}{2}\right)\left(1 + O\left(\frac{\log n}{\sqrt{n}}\right)\right)\\ &=\; \sum_{j=1}^{10\log n\sqrt{n}} \exp\left(-\frac{\pi}{\sqrt{6}}\,jt\right)\left(1 + O\left(\frac{\log n}{\sqrt{n}}\right)\right)\\ &=\; \frac{1}{\exp\left(\frac{\pi}{\sqrt{6}}t\right) - 1} + O\left(\frac{\log n}{\sqrt{n}}\right),\end{aligned}$$

where the last step is by geometric series formula. This proves the lemma. □

Proof of Theorem 3. Put $c = \pi/\sqrt{6}$. Recall that the height of the Young diagram at $x = t$ is

$$H_t \;=\; \frac{1}{\sqrt{n}} \sum_{i=t\sqrt{n}}^{n} X_i.$$

Thus, linearity of expectation gives

$$\begin{aligned}\mathbb{E}[H_{t,n}] \;&=\; \frac{1}{\sqrt{n}} \sum_{i=t\sqrt{n}}^{n} \mathbb{E}[X_i]\\ &=\; \frac{1}{\sqrt{n}} \sum_{i=t\sqrt{n}}^{n} \frac{1}{\exp\left(\frac{ci}{\sqrt{n}}\right) - 1}\left(1 + O\left(\frac{\log n}{\sqrt{n}}\right)\right).\end{aligned}$$

We will split the sum at $i = 10\sqrt{n}\log n$. From $i > 10\sqrt{n}\log n$ onward, the terms have naive bound

$$\frac{1}{\sqrt{n}} \sum_{i=10\sqrt{n}\log n}^{n} \frac{1}{\exp\left(\frac{ci}{\sqrt{n}}\right) - 1} \;\ll\; \sum_{i=10\sqrt{n}\log n}^{n} \frac{1}{n^{10} - 1} \;\ll\; n^{-9}.$$

For $i \le 10\sqrt{n}\log n$, we may bound the sum by integrals. Since the function $x \mapsto \frac{1}{e^{cx}-1}$ is decreasing, we get that

$$\int_{t+1/\sqrt{n}}^{10\log n + 1/\sqrt{n}} \frac{1}{e^{cx} - 1}\,dx \;\le\; \frac{1}{\sqrt{n}} \sum_{i=t\sqrt{n}}^{10\sqrt{n}\log n} \frac{1}{\exp\left(\frac{ci}{\sqrt{n}}\right) - 1} \;\le\; \int_{t}^{10\log n} \frac{1}{e^{cx} - 1}\,dx.$$

The difference between our upper and lower bound is $O\left(\frac{1}{\sqrt{n}}\right)$ because it can be written as the difference of two integrals each over an interval of length $\frac{1}{\sqrt{n}}$. Therefore, we get that

$$\begin{aligned}
\mathbb{E}[H_t] &= \frac{1}{\sqrt{n}} \sum_{i=t\sqrt{n}}^{n} \frac{1}{\exp\left(\frac{ci}{\sqrt{n}}\right)-1}\left(1+O\left(\frac{\log n}{\sqrt{n}}\right)\right) \\
&= \int_t^{10\log n} \frac{1}{\exp(cx)-1}\,dx + O\left(\frac{\log n}{\sqrt{n}}\right) \\
&= \int_t^{\infty} \frac{1}{\exp(cx)-1}\,dx - \int_{10\log n}^{\infty} \frac{1}{\exp(cx)-1}\,dx + O\left(\frac{\log n}{\sqrt{n}}\right) \\
&= \int_t^{\infty} \frac{1}{\exp(cx)-1}\,dx + O\left(\int_{10\log n}^{\infty} e^{-cx}\,dx\right) + O\left(\frac{\log n}{\sqrt{n}}\right) \\
&= \int_t^{\infty} \frac{1}{\exp(cx)-1}\,dx + O\left(\frac{\log n}{\sqrt{n}}\right) \\
&= t - \frac{1}{c}\log(\exp(ct)-1) + O\left(\frac{\log n}{\sqrt{n}}\right) \\
&= t - \frac{\sqrt{6}}{\pi}\log\left(\exp\left(\frac{\pi}{\sqrt{6}}t\right)-1\right) + O\left(\frac{\log n}{\sqrt{n}}\right).
\end{aligned}$$

□

4. Outline of Concentration Proof

We start setting up for the proof of our concentration result Theorem 4, reducing it to two key inputs (Proposition 15 and Proposition 18), which we handle in the following Section 5 and Section 6.

From now on, we assume that n is sufficiently large. Take a uniform random partition $\lambda = (\lambda_1, \ldots, \lambda_r)$ of n, and let X_i be the number of i's in that partition (i.e., the number of i's that appear among $\lambda_1, \ldots, \lambda_r$.) We wish to prove the concentration of

$$H_t = \frac{1}{\sqrt{n}} \sum_{i=t\sqrt{n}}^{n} X_i.$$

A naive attempt is to apply Azuma's inequality (Theorem 12) on the Doob Martingale

$$\boldsymbol{Z_{n-q}^{\mathrm{bad}}} = \mathbb{E}\left[\sum_{i=t\sqrt{n}}^{n} X_i \;\middle|\; X_{q+1}, X_{q+2}, \ldots, X_n\right].$$

Computing Z_{n-q}^{bad} is fairly tractable. However, there are two obstructions for this martingale to satisfy (1).

(1) First, if X_q is very large, then Z_{n-q}^{bad} (the expected value evaluated when X_j is revealed) will be much smaller than Z_{n-q+1}^{bad} (the expected value evaluated when X_j is not revealed), since the partition skews away from smaller parts that the number of parts is smaller. This obstructs (1).

(2) Second, we note that $\sum_{i=1}^{n} iX_i = n$. Thus, if it is revealed that $\sum_{i=q+1}^{n} iX_i$ is large, then the $X_1, X_2, \ldots, X_q$ are small. This allows the difference $|Z_{n-q}^{\mathrm{bad}} - Z_{n-q+1}^{\mathrm{bad}}|$ to fluctuate and violates (1).

In order to fix this issue, we define

$$\boldsymbol{R_q} \;=\; n - \sum_{i=q+1}^{n} iX_i,$$

be the remainder of the partition after $X_{q+1}, X_{q+2}, \ldots, X_n$ are revealed. Let $\boldsymbol{E}$ be the event that both of the following holds:

(1) we have that for all i,

$$X_i \;<\; \frac{20\sqrt{n}\log n}{i}.$$

(2) we have

$$R_{t\sqrt{n}} \;<\; (1-c_t)n$$

for some constant $c_t > 0$ specified in Proposition 20.

The following proposition allows us to avoid these events. Its proof is wholly unrelated to this section, and is therefore proved in Section 5.

Proposition 15. *We have $\mathbb{P}[E] \geq 1 - \frac{1}{n^9}$ for all sufficiently large n.*

We will apply Azuma's inequality on the martingale

$$\boldsymbol{Z_{n-q}^{\text{good}}} \;=\; \mathbb{E}\left[\sum_{i=t\sqrt{n}}^{n} X_i \;\middle|\; X_{q+1}, X_{q+2}, \ldots, X_n, E\right].$$

4.1. **Comparing bad and good martingales.** To evaluate the value of Z_{n-q}^{good}, we do this by computing Z_{n-q}^{bad} (which is easier to compute) and show that it does not deviate too much from Z_{n-q}^{good}. To relate Z_{n-q}^{bad} and Z_{n-q}^{good}, we prove the following easy lemma.

Lemma 16. *Let X be a random variable outputting a number in $[0, M]$ and E be an event happening with probability $1-p$. Then we have*

$$\mathbb{E}[X] - pM \;\leq\; \mathbb{E}[X \mid E] \;\leq\; \mathbb{E}[X] + pM$$

Proof. We have

$$\mathbb{E}[X] \;=\; (1-p)\cdot\mathbb{E}[X \mid E] \;+\; p\cdot\mathbb{E}[X \mid \text{not } E].$$

To get the second inequality, note that $\mathbb{E}[X \mid \text{not } E] \geq 0$, so the inequality above turns $\mathbb{E}[X] \geq (1-p)\mathbb{E}[X \mid E]$, which implies that

$$\mathbb{E}[X \mid E] - \mathbb{E}[X] \;\leq\; p\,\mathbb{E}[X \mid E] \leq pM,$$

as desired.

To get the first inequality, note that $\mathbb{E}[X \mid \text{not } E] < M$, so we have that

$$\mathbb{E}[X] \;\leq\; (1-p)\cdot\mathbb{E}[X \mid E] + pM \;\leq\; \mathbb{E}[X \mid E] + pM,$$

implying the conclusion. □

Corollary 17. *We have*

$$\left|Z_{n-q}^{bad} - Z_{n-q}^{good}\right| \;\leq\; \frac{1}{n^7}.$$

Proof. Follows from applying the previous lemma with $X = \sum_{i=t\sqrt{n}}^{n} X_i$ conditioned on the values of $X_{q+1}, \ldots, X_n$, $p = n^{-9}$, and $M = n^2$. □

4.2. **Applying Azuma's Inequality.** The following proposition verifies the condition needed to apply Azuma's inequality. Once again, the idea for this proof is unrelated to this section, and is therefore proved in Section 6.

Proposition 18. *We have* $|Z^{good}_{n-q} - Z^{good}_{n-q+1}| = O(n^{1/3+\varepsilon})$.

Using this, we complete the proof of Theorem 4.

Proof of Theorem 4 assuming Proposition 15 and Proposition 18. Let K be a constant such that $|Z^{\text{good}}_{n-q} - Z^{\text{good}}_{n-q+1}| < Kn^{1/3+\varepsilon/2}$. Thus, by Azuma's inequality (Theorem 12), we get that

$$\mathbb{P}\big[|Z^{\text{good}}_n - Z^{\text{good}}_0| \geq 2\sqrt{n}\cdot\lambda\big] \;\leq\; \exp\left(-\frac{(2\sqrt{n}\cdot\lambda)^2}{2K^2n^{2/3+\varepsilon}}\right) \;\leq\; \exp\left(-\frac{2\lambda^2}{K^2n^{-1/3+\varepsilon}}\right).$$

Next, we note that $Z^{\text{bad}}_n = \sqrt{n}\,H_t$ and $Z^{\text{bad}}_0 = \sqrt{n}\mathbb{E}[H_t]$. Therefore,

$$\begin{aligned} |Z^{\text{good}}_n - Z^{\text{good}}_0| \;&\geq\; \sqrt{n}\,|H_t - \mathbb{E}[H_t]| - |Z^{\text{good}}_n - Z^{\text{bad}}_n| - |Z^{\text{good}}_0 - Z^{\text{bad}}_0| \\ &\geq\; \sqrt{n}\,|H_t - \mathbb{E}[H_t]| - \frac{2}{n^7} \qquad \text{(by Corollary 17)} \end{aligned}$$

Thus, since $\lambda \geq \frac{1}{n^6}$, we get that the event $|H_t - \mathbb{E}[H_t]| \geq \lambda$ implies $|Z^{\text{good}}_n - Z^{\text{good}}_0| \geq 2\sqrt{n}\cdot\lambda$. Hence, we have that

$$\mathbb{P}\big[|H_t - \mathbb{E}[H_t]| \geq \lambda\big] \geq \exp\left(-\frac{2\lambda^2}{K^2n^{-1/3+\varepsilon}}\right),$$

which implies the result after setting $C = \frac{K^2}{2}$. □

Thus, we have now reduced Theorem 4 to proving Proposition 15 and Proposition 18, which we take Section 5 and Section 6 to do so.

5. Avoiding Two Bad Events

The goal of this section is to prove Proposition 15, which states that the probability that E happens is at least $1 - \frac{1}{n^9}$. To do this, we have to upper bound the probabilities that the bullet points (1) and (2) (in the definition of E in Section 4) are violated. Proposition 19 deals with (1), while Proposition 20 deals with (2).

Proposition 19. *There is some constant D such that for all i and n we have*

$$\mathbb{P}\left[X_i > \frac{20\sqrt{n}\,\log n}{i}\right] \;<\; \frac{D}{n^{11}}.$$

Proof. The key observation is that there is an injection from partitions of n with at least $20\frac{\sqrt{n}}{i}\log n$ parts of size i to partitions of $n - i\lceil 20\sqrt{n}\log n/i\rceil$ by removing exactly $\lceil 20\frac{\sqrt{n}}{i}\log n\rceil$ occurrences of i. Therefore, we conclude

$$\begin{aligned} \mathbb{P}\left(X_i > 20\frac{\sqrt{n}}{i}\log n\right) \;&\leq\; \frac{p(n - i\lceil 20\sqrt{n}\log n/i\rceil)}{p(n)} \\ &\ll\; \frac{\frac{1}{n-20\sqrt{n}\log n}\exp\left(\pi\sqrt{\frac{2}{3}(n-20\sqrt{n}\log n)}\right)}{\frac{1}{n}\exp\left(\pi\sqrt{\frac{2}{3}n}\right)} \\ &\ll\; \exp\left(\pi\sqrt{\frac{2}{3}}\left(\sqrt{n-20\sqrt{n}\log n} - \sqrt{n}\right)\right) \end{aligned}$$

$$\leq \exp\left(\pi\sqrt{\frac{2}{3}}\left(-9\log n\right)\right)$$
$$\ll \frac{1}{n^{11}},$$

where we use that n is sufficiently large throughout. □

Proposition 20. *Fix $t > 0$. There are positive constants $c_t, D_t > 0$ such that*

$$\mathbb{P}\left[\sum_{i=t\sqrt{n}}^{n} iX_i \geq (1-c_t)n\right] < \frac{D_t}{n^{11}}.$$

for all n such that D_t is uniformly bounded on compact subsets of $(0,\infty)$.

Proof. From Proposition 19, with very high probability, we know that $X_q = 0$ for all $q > 20\sqrt{n}\log n$. Thus, with very high probability $i > \frac{\sqrt{n}}{20\log n}$.

Observe that partitions of m into parts of size $> t\sqrt{n}$ with exactly j parts can be bijected with partitions of $m - j\lfloor t\sqrt{n}\rfloor$ into exactly j parts.

Thus,

$$\begin{aligned}
\text{LHS} &= \sum_{m=(1-c_t)n}^{n} \mathbb{P}\left[\sum_{i=t\sqrt{n}}^{n} iX_i = m\right] \\
&= \sum_{m=(1-c_t)n}^{n} \frac{p(n-m, t\sqrt{n})}{p(n)} \sum_{j=\frac{\sqrt{n}}{20\log n}}^{\sqrt{n}/t} p(m - j\lfloor t\sqrt{n}\rfloor, j) - p(m - j\lfloor t\sqrt{n}\rfloor, j-1) \\
&\leq n\frac{p(c_t n, t\sqrt{n})}{p(n)} \sum_{j=\frac{\sqrt{n}}{20\log n}}^{\sqrt{n}/t} p(n - j\lfloor t\sqrt{n}\rfloor, j) \\
&\leq n^2\frac{p(c_t n)}{p(n)} p(n, \sqrt{n}/t) \ll n^3 \exp\left(\left(g\left(\tfrac{1}{t}\right) + \sqrt{c_t}\,\pi\sqrt{\tfrac{2}{3}} - \pi\sqrt{\tfrac{2}{3}}\right)\cdot\sqrt{n}\right).
\end{aligned}$$

Now, since g is strictly increasing and limits to $\pi\sqrt{2/3}$ (by Proposition 9), we know that $g(1/t) < \pi\sqrt{2/3}$, and thus we can find some positive $c_t \ll 1$ such that $g(1/t) + \sqrt{c_t}\pi\sqrt{2/3} - \pi\sqrt{2/3} < 0$. □

To prove Proposition 15, we simply take a union bound of the Proposition 19 (across all i) and Proposition 20.

6. Computing Martingales

The goal of this section is to prove Proposition 18, which upper bounds the difference $|Z^{\text{good}}_{n-q} - Z^{\text{good}}_{n-q+1}|$. To do this, we start out by computing Z^{bad}_{n-q}. Then, we use Corollary 17 to link the bad martingale to the good martingale, allowing us to finish the proof.

Since Z^{bad}_{n-q} is the sum of conditional expected value $\mathbb{E}[X_i \mid X_{q+1}, X_{q+2}, \ldots, X_n]$, we begin by finding a closed form for this expression.

Lemma 21. *We have*

$$\mathbb{E}[X_i \mid X_{q+1}, X_{q+2}, \ldots, X_n] = \frac{1}{p(R_q, q)} \sum_{j=1}^{\infty} p(R_q - ji, q). \tag{3}$$

Proof. This follows from the exact same idea as the combinatorial proof of Lemma 13. □

We now estimate the right hand side of (3). To do this, recall the functions f and g from Theorem 7. Recall also that $h(u) = ug(1/u)$.

Lemma 22. *(a) If q and i are integers such that $\sqrt{n}\log^2 n < i < q$, then*

$$\mathbb{E}[X_i \mid X_{q+1}, X_{q+2}, \ldots, X_n] \;<\; \frac{1}{n^2}.$$

(b) Let $\varepsilon > 0$. Uniformly for all q and i such that $q > i > t\sqrt{n}$ and $R_q > c_t n$, we have

$$\mathbb{E}[X_i \mid X_{q+1}, X_{q+2}, \ldots, X_n] \;=\; \frac{1 + O(n^{-1/6+\varepsilon})}{\exp\left(\frac{\alpha i}{\sqrt{n}}\right) - 1},$$

where $\alpha = \sqrt{\frac{n}{R_q}} h'\left(\frac{\sqrt{R_q}}{q}\right)$, which is bounded.

Proof. We define $F(x) = \frac{1}{x} f\left(\frac{q}{\sqrt{x}}\right)$. From Theorem 7, we note that uniformly for all $x > c_t n$, we have

$$\begin{aligned} p(x,q) \;&=\; (1 + O(n^{-1/6+\varepsilon}))F(x)\exp\left(g\left(\frac{q}{\sqrt{x}}\right)\sqrt{x}\right) \\ &=\; (1 + O(n^{-1/6+\varepsilon}))F(x)\exp\left(q \cdot h\left(\frac{\sqrt{x}}{q}\right)\right) \end{aligned}$$

Thus, the desired expected value is

$$\begin{aligned} &\mathbb{E}[X_i \mid X_{q+1}, \ldots, X_n] \\ &\quad=\; (1 + O(n^{-1/6+\varepsilon}))\sum_{j=1}^{\infty} \frac{F(R_q - ji)}{F(R_q)} \exp\left(q \cdot \left(h\left(\frac{\sqrt{R_q - ji}}{q}\right) - h\left(\frac{\sqrt{R_q}}{q}\right)\right)\right). \end{aligned} \tag{4}$$

We wish to use Taylor's theorem to approximate the difference between two h's in the exponent. To do that, we first approximate their difference.

$$\begin{aligned} \frac{\sqrt{R_q}}{q} - \frac{\sqrt{R_q - ji}}{q} \;&=\; \frac{\sqrt{R_q}}{q}\left(1 - \sqrt{1 - \frac{ji}{R_q}}\right) \\ &=\; \frac{\sqrt{R_q}}{q}\left(\frac{ji}{R_q} + O\left(\frac{j^2 i^2}{R_q^2}\right)\right) \\ &=\; \frac{ji}{q\sqrt{R_q}} + O\left(\frac{j^2 i^2}{q R_q^{3/2}}\right). \end{aligned} \tag{5}$$

Therefore, by Taylor's theorem, we get that

$$\begin{aligned} h\left(\frac{\sqrt{R_q - ji}}{q}\right) - h\left(\frac{\sqrt{R_q}}{q}\right) \;&=\; -\frac{ji}{2q\sqrt{R_q}} \cdot h'\left(\frac{R_q}{q^2}\right) + O\left(\frac{j^2 i^2}{q R_q^{3/2}}\right) + O\left(\frac{j^2 i^2}{2q^2 R_q}\right) \\ &=\; -\frac{ji}{2q\sqrt{R_q}} \cdot h'\left(\frac{R_q}{q^2}\right) + O\left(\frac{j^2 i^2}{q R_q^{3/2}}\right). \end{aligned} \tag{6}$$

In the second equality, the first big-O comes from the error in (5), and we use Proposition 9 to say that h' is bounded. The second big-O comes from the error term in the Taylor expansion,

which is valid because by Proposition 9, second derivative h'' is bounded. The third equality follows from $R_q > c_t n > \frac{c_t}{t^2} q^2$, so $R_q = \Omega(q^2)$.

We note that $t < q/\sqrt{R_q} < 1/c_t$, so $h'(R_q/q^2)$ is bounded. Thus, if $ji \ll n$, then the error term is dominated by the first term.

Next, we note that the terms when $ji > \sqrt{n}\log^2 n$ do not contribute to the sum. To see this, note that if $j = \lceil \sqrt{n}\log^2 n/i \rceil$, then

$$\begin{aligned} q\left(h\left(\frac{\sqrt{R_q - ji}}{q}\right) - h\left(\frac{\sqrt{R_q}}{q}\right)\right) &\leq -\frac{ji}{3\sqrt{R_q}} h'\left(\frac{\sqrt{R_q}}{q}\right) \\ &\leq -\frac{\sqrt{n}\log^2 n}{2\cdot\sqrt{n}} \cdot h'\left(\frac{\sqrt{R_q}}{q}\right) \leq -d_t \log^2 n, \end{aligned}$$

for some constant d_t. In particular, since f is bounded, we get that the term at $j = \lceil \sqrt{n}\log^2 n/i \rceil$ is at most $O(\exp(-d_t \log^2 n))$. Since terms with higher j are even smaller, the sum in range $j > \sqrt{n}\log^2 n/i$ is smaller than $O(n\exp(-d_t\log^2 n))$.

We are now ready to deduce each part of the lemma.

(a) If $i > \sqrt{n}\log^2 n$, then $ji > \sqrt{n}\log^2 n$ for all $j \geq 1$, so the desired expected value is $O(n\exp(-d_t\log^2 n)) \ll O(1/n^2)$, proving (a).

(b) If $i \leq \sqrt{n}\log^2 n$, then it suffices to consider the sum in range $ji \leq \sqrt{n}\log^2 n$, in which case we have that the exponent is

$$\begin{aligned} q\left(h\left(\frac{\sqrt{R_q - ji}}{q}\right) - h\left(\frac{\sqrt{R_q}}{q}\right)\right) &= -\frac{ji}{2\sqrt{R_q}} h'\left(\frac{\sqrt{R_q}}{q}\right) + O\left(\frac{j^2 i^2}{R_q^{3/2}}\right) \\ &= -\frac{\alpha ij}{\sqrt{n}} + O\left(\frac{\log^4 n}{\sqrt{n}}\right), \end{aligned}$$

so we have that

$$\exp\left(q\cdot\left(h\left(\frac{\sqrt{R_q - ji}}{q}\right) - h\left(\frac{\sqrt{R_q}}{q}\right)\right)\right) = \left(1 + O\left(\frac{\log^4 n}{\sqrt{n}}\right)\right)\cdot \exp\left(-\frac{\alpha ij}{\sqrt{n}}\right). \tag{7}$$

Next, we show that $\frac{F(R_q - ji)}{F(R_q)}$ is close to 1. To do this, we note that

$$\frac{F(R_q - ji)}{F(R_q)} = \frac{R_q}{R_q - ji} \cdot \frac{f\left(\frac{q}{\sqrt{R_q - ji}}\right)}{f\left(\frac{q}{\sqrt{R_q}}\right)}.$$

The first factor is approximately $1 + O\left(\frac{ij}{R_q}\right) = 1 + O\left(\frac{\log^2 n}{\sqrt{n}}\right)$. For the second factor, we note that

$$\begin{aligned} \frac{q}{\sqrt{R_q - ji}} - \frac{q}{\sqrt{R_q}} &= \frac{qij}{\sqrt{R_q(R_q - ji)}(\sqrt{R_q} + \sqrt{R_q - ji})} \\ &= O\left(\frac{qij}{R_q^{3/2}}\right) = O\left(\frac{\log^2 n}{\sqrt{n}}\right), \end{aligned}$$

so by Proposition 9 that f is Lipschitz, we get that

$$f\left(\frac{q}{\sqrt{R_q - ji}}\right) - f\left(\frac{q}{\sqrt{R_q}}\right) \;=\; O\left(\frac{\log^2 n}{\sqrt{n}}\right),$$

and since f is bounded above, we get that the second factor is also $1 + O\left(\frac{\log^2 n}{\sqrt{n}}\right)$. Hence, we get that

$$\frac{F(R_q - ji)}{F(R_q)} \;=\; 1 + O\left(\frac{\log^2 n}{\sqrt{n}}\right). \tag{8}$$

Finally, inserting (7) and (8) into (4), we get that

$$\mathbb{E}[X_i \mid X_{q+1}, \ldots, X_n] \;=\; (1 + O(n^{-1/6+\varepsilon})) \sum_{j=1}^{\infty} \exp\left(-\frac{\alpha i j}{\sqrt{n}}\right) \;=\; \frac{1 + O(n^{-1/6+\varepsilon})}{\exp\left(\frac{\alpha i}{\sqrt{n}}\right) - 1},$$

where we use geometric series in the last step. This completes the proof of (b). □

Finally, we sum the estimates in the Lemma 22 to get Z^{bad}_{n-q}.

Proposition 23. *We have*

$$Z^{bad}_{n-q} \;=\; \sqrt{n}\left(t - \frac{1}{\alpha}\log(e^{\alpha t} - 1)\right) + O(n^{1/3+\varepsilon}),$$

where α is as in Lemma 22 (b).

Proof. By linearity of expectation, Z^{bad}_{n-q} is the sum $\mathbb{E}[X_i \mid X_{q+1}, \ldots, X_n]$ as i ranges from $t\sqrt{n}$ to n. By Lemma 22 (a), the terms for $i > \sqrt{n}\log^2 n$ contributes at most $n \cdot \frac{1}{n^2} = \frac{1}{n}$, which is less than our error term. Thus, we have that

$$Z^{\text{bad}}_{n-q} \;=\; \left(\sum_{i=t\sqrt{n}}^{\sqrt{n}\log^2 n} \frac{1}{\exp\left(\frac{\alpha i}{\sqrt{n}}\right) - 1}\right) + O(n^{1/3+\varepsilon}) \tag{9}$$

(where we applied Lemma 22 (b) on $\varepsilon/2$ instead of ε to account for the extra factor of log). It remains to estimate the sum in the inside. To do this, we first note that since $\frac{\sqrt{R_q}}{q} > \frac{c_t}{t}$, we get that α is uniformly bounded by some $\delta > 0$. Thus, in our estimates, we may treat $\frac{1}{\alpha}$ as constant in the big-O. Next, we note that since the function $x \mapsto \frac{1}{e^{\alpha x}-1}$ is decreasing in $[t, \infty)$, we get that

$$\sqrt{n}\int_{t+1/\sqrt{n}}^{\log n + 1/\sqrt{n}} \frac{1}{e^{\alpha x} - 1}\,dx \;\le\; \sum_{i=t\sqrt{n}}^{\sqrt{n}\log^2 n} \frac{1}{\exp\left(\frac{\alpha i}{\sqrt{n}}\right) - 1} \;\le\; \sqrt{n}\int_{t}^{\log^2 n} \frac{1}{e^{\alpha x} - 1}\,dx.$$

The difference between the upper and lower bound is $O(1)$, since it is $\sqrt{n}$ times an integral over an interval of length $O\left(\frac{1}{\sqrt{n}}\right)$. Therefore, we get that

$$\begin{aligned}\sum_{i=t\sqrt{n}}^{\sqrt{n}\log^2 n} \frac{1}{\exp\left(\frac{\alpha i}{\sqrt{n}}\right) - 1} \;&=\; \int_{t}^{\log^2 n} \frac{1}{e^{\alpha x} - 1}\,dx + O(1) \\ &=\; \sqrt{n}\int_{t}^{\infty} \frac{1}{e^{\alpha x} - 1} - \sqrt{n}\int_{\log^2 n}^{\infty} \frac{1}{e^{\alpha x} - 1} + O(1)\end{aligned}$$

$$
\begin{aligned}
&= \sqrt{n}\int_t^\infty \frac{1}{e^{\alpha x}-1} + O\left(\sqrt{n}\int_{\log^2 n}^\infty e^{-\alpha x}\right) + O(1)\\
&= \sqrt{n}\int_t^\infty \frac{1}{e^{\alpha x}-1} + O\left(\frac{\sqrt{n}}{\alpha}\, e^{-\alpha \log^2 n}\right) + O(1)\\
&= \sqrt{n}\int_t^\infty \frac{1}{e^{\alpha x}-1} + O(1)\\
&= \sqrt{n}\left(t - \frac{1}{\alpha}\log(e^{\alpha t}-1)\right) + O(1).
\end{aligned}
$$

Combining this with (9) yields the result. □

Next, we apply Corollary 17 to obtain an estimate of Z_{n-q}^{good}.

Corollary 24. *We have*

$$
Z_{n-q}^{good} = \sqrt{n}\left(t - \frac{1}{\alpha}\log(e^{\alpha t}-1)\right) + O(n^{1/3+\varepsilon}),
$$

where α is as in Lemma 22 (b).

Proof. Follows directly from Proposition 23 and Corollary 17. □

We are now ready to prove Proposition 18, which we reproduce below.

Proposition. *We have $|Z_{n-q}^{good} - Z_{n-q+1}^{good}| = O(n^{1/3+\varepsilon})$.*

Proof . Let $\alpha' = \sqrt{\frac{n}{R_{q-1}}}h'\left(\frac{\sqrt{R_{q-1}}}{q-1}\right)$ be the constant obtained by replacing q with $q-1$ in Lemma 22. Then by Corollary 24, we get that

$$
Z_{n-q}^{\text{good}} - Z_{n-q+1}^{\text{good}} = \sqrt{n}\left(\frac{1}{\alpha'}\log(e^{\alpha' t}-1) - \frac{1}{\alpha}\log(e^{\alpha t}-1)\right) + O(n^{1/3+\varepsilon}).
$$

We now have to bound the difference between α and α'. Since $q > t\sqrt{n}$ and $\sqrt{R_q} < \sqrt{n}$, we get that $\frac{\sqrt{R_q}}{q} < \frac{1}{t}$, and similarly, $\frac{\sqrt{R_{q-1}}}{q-1} < \frac{1}{t}$, so $h'\left(\frac{\sqrt{R_q}}{q}\right)$ and $h'\left(\frac{\sqrt{R_{q-1}}}{q-1}\right)$ are bounded by some constant K (depending on t). Combining with $R_q > c_t n$ (holds since E holds), we get that

$$
|\alpha - \alpha'| < K\left|\sqrt{\frac{n}{R_q}} - \sqrt{\frac{n}{R_{q-1}}}\right| + \frac{1}{\sqrt{c_t}}\left|h'\left(\frac{\sqrt{R_q}}{q}\right) - h'\left(\frac{\sqrt{R_{q-1}}}{q-1}\right)\right|.
$$

We now analyze each of the two terms. For the first term, note that

$$
\begin{aligned}
\left|\sqrt{\frac{n}{R_q}} - \sqrt{\frac{n}{R_{q-1}}}\right| &= \sqrt{n}\cdot\left|\frac{R_q - R_{q-1}}{\sqrt{R_q R_{q-1}}(\sqrt{R_q}+\sqrt{R_{q-1}})}\right|\\
&\le \sqrt{n}\cdot\frac{qX_q}{2(c_t n)^{3/2}}\\
&\le \sqrt{n}\cdot\frac{20\sqrt{n}\log n}{2(c_t n)^{3/2}} \qquad (E \text{ holds, so } X_q \le \tfrac{20\sqrt{n}\log n}{q})\\
&\le \frac{10}{c_t^{3/2}}\cdot\frac{\log n}{\sqrt{n}} = O\left(\frac{\log n}{\sqrt{n}}\right).
\end{aligned}
$$

For the second term, we note that

$$\begin{aligned}
\left|\frac{\sqrt{R_q}}{q} - \frac{\sqrt{R_{q-1}}}{q-1}\right| &\leq \left|\frac{\sqrt{R_q} - \sqrt{R_{q-1}}}{q}\right| + \frac{\sqrt{R_q}}{q(q-1)} \\
&= \left|\frac{R_q - R_{q-1}}{q(\sqrt{R_q} + \sqrt{R_{q-1}})}\right| + \frac{\sqrt{R_q}}{q(q-1)} \\
&\leq \frac{X_q}{2\sqrt{c_t n}} + \frac{\sqrt{n}}{t^2 n} \\
&\leq \frac{20\sqrt{n}\log n}{2q\sqrt{c_t n}} + \frac{1}{t^2} \qquad (E \text{ holds, so } X_q \leq \tfrac{20\sqrt{n}\log n}{q}) \\
&\leq \frac{10\log n}{t\sqrt{c_t}} + \frac{1}{t^2} = O\left(\frac{\log n}{\sqrt{n}}\right).
\end{aligned}$$

Thus, we deduce that $|\alpha - \alpha'| = O(\log n/\sqrt{n})$.

Finally, note that the function

$$\phi(x) \;=\; \frac{1}{x}\log(e^{xt} - 1)$$

has the first derivative

$$\phi'(x) \;=\; \frac{te^{tx}}{x(e^{tx}-1)} - \frac{\log(e^{tx}-1)}{x^2},$$

which is continuous in $[\delta, \infty)$ and converges to 0 as $x \to \infty$ for any fixed $\delta > 0$, so by compactness argument, ϕ' is bounded in $[\delta, \infty)$. Since α is bounded, we get that there exists some constant M, only depending on t, such that $|\phi(\alpha) - \phi(\alpha')| < M|\alpha - \alpha'|$. This implies that

$$Z^{\text{good}}_{n-q} - Z^{\text{good}}_{n-q+1} \;=\; \sqrt{n}\,|\phi(\alpha) - \phi(\alpha')| + O(n^{1/3+\varepsilon}) \;=\; O(n^{1/3+\varepsilon}),$$

as desired. □

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Work distribution for the first author:

- coded the Monte Carlo simulation of random partitions and the numerical calculation of the expected height (which was very helpful in figuring out the limiting curve);
- worked out the details and wrote up proofs in Section 3, Section 5, and Section A.

Work distribution for the second author:

- devised the strategy for the proof of Theorem 4.
- worked out the details and wrote up proofs in Section 4 and Section 6.

Both authors worked together to figure out the limiting curve and to write Section 1 and Section 2.

Appendix A. Proof of Proposition 9

For convenience, we first recall the definition of all functions. First, $v = v(u) : (0,\infty) \to (0,\infty)$ is defined implicitly by

$$u^2 \;=\; \frac{v^2}{\int_0^v \frac{x}{e^x-1}\,dx}.$$

For simplicity of notation, we will put

$$c(v) \;\coloneqq\; \int_0^v \frac{x}{e^x-1}\,dx \quad \text{so that} \quad u^2 \;=\; \frac{v^2}{c(v)}. \tag{10}$$

Then f and g are defined by

$$\begin{aligned} f(u) \;&=\; \frac{v}{2^{3/2}\pi u}\left(1-e^{-v}-\tfrac{1}{2}u^2e^{-v}\right)^{-1/2} \\ g(u) \;&=\; \frac{2v}{u} - u\log(1-e^{-v}). \end{aligned}$$

We reproduce Proposition 9 below.

Proposition. *The following properties of f and g hold.*

(a) f is bounded, and there exist a constant C such that f is C-Lipschitz.
(b) The function g is strictly increasing and $\lim_{u\to\infty} g(u) = \pi\sqrt{2/3}$.
(c) Put $h(t) = t\,g(1/t)$. Then
- *h is strictly increasing*
- *h has first and second derivatives that are both continuous and bounded.*

A.1. **Smoothness of v away from 0.** To show that v is smooth in u, we first recall the following consequence of the inverse function theorem.

Lemma 25. *Let $a(b(x)) = x$ be functions such that a is smooth and a' is non-zero in some neighborhood of x_0. Then in some neighborhood of $b(x_0)$, b is smooth.*

Lemma 26. *The function v is smooth in u on any compact subset of $(0,\infty)$.*

Proof. We apply Lemma 25 and think of b mapping u to v and a mapping v to u. Specifically, for any $v \in (0,\infty)$, we have

$$a(v) \;=\; \frac{v}{\sqrt{c(v)}}.$$

Here, the right side is smooth since $x \mapsto \sqrt{x}$ is smooth away from 0, and the integral is smooth because it is integrating something smooth to and from smooth endpoints. It thus remains to show that $a'(v) \neq 0$ for all $v \in (0,\infty)$. Observe that $a'(v) = 0$ if and only if the following expression is 0

$$\sqrt{c(v)} - v\cdot\frac{d}{dv}\sqrt{c(v)} \;=\; \sqrt{c(v)} - v\cdot\frac{c'(v)}{2\sqrt{c(v)}} \;=\; \sqrt{c(v)} - \frac{v^2}{2(e^v-1)}\cdot\frac{1}{\sqrt{c(v)}}$$

Now, clearing the denominators means that we would need $c(v) = d(v)$ for

$$d(v) \;\coloneqq\; \frac{v^2}{2(e^v-1)}.$$

Now, we have $c(0) = 0$ and $d(v) \to 0$ as $v \to 0$. Moreover, taking the first derivative gives

$$c'(v) \;=\; \frac{v}{e^v-1} \quad \text{and} \quad d'(v) \;=\; \frac{v}{e^v-1} - \frac{v^2e^v}{2(e^v-1)^2},$$

and thus $c(v) > d(v)$ for any positive v. □

A.2. **Derivatives of v near 0 and ∞.** Having controlled the behavior of v near any point other than 0 or ∞, we now focus on the behavior near 0 and ∞. We begin by giving asymptotic expansion of those values itself.

Lemma 27 (Behavior near ∞)**.** *As $u \to \infty$, or, equivalently, as $v \to \infty$, we have*

(a) $c(v) = \frac{\pi^2}{6} + O(ve^{-v})$;
(b) $\frac{v}{u} = \frac{\pi}{\sqrt{6}} + O(ve^{-v}) = \frac{\pi}{\sqrt{6}} + O(ue^{-u})$;
(c) $\frac{dv}{du} = \frac{\pi}{\sqrt{6}} + O(v^3e^{-v})$; *and*
(d) $\frac{d^2v}{du^2} = O(v^3e^{-v})$.

Proof. (a) We first evaluate $\lim_{v\to\infty} c(v)$. To do this, we expand the relevant integral into infinite series:

$$\int_0^\infty \frac{x\,dx}{e^x - 1} \;=\; \int_0^\infty \sum_{n=1}^\infty xe^{-nx}\,dx \;=\; \sum_{n=1}^\infty \int_0^\infty xe^{-nx}\,dx \;=\; \sum_{n=1}^\infty \frac{1}{n^2} \;=\; \frac{\pi^2}{6},$$

where we use dominated convergence theorem to justify swapping sums and integrals.

As $v \to \infty$, we have

$$0 \;<\; \frac{\pi^2}{6} - c(v) \;=\; \int_v^\infty \frac{x\,dx}{e^x - 1} \le \int_v^\infty 2xe^{-x}\,dx = O(ve^{-v}).$$

(b) From (a), we immediately get that

$$\frac{v}{u} \;=\; \frac{\pi}{\sqrt{6}} + O(ve^{-v}) = \frac{\pi}{\sqrt{6}} + O(ue^{-u}).$$

(c) Differentiating (10) with respect to v gives

$$2u\frac{du}{dv} \;=\; \frac{2v}{c(v)} - \frac{v^2}{c(v)^2}\frac{v}{e^v - 1} \tag{11}$$

for all u. Then, using (a) and (b) gives that as $v \to \infty$

$$\frac{du}{dv} \;=\; \frac{\sqrt{6}}{\pi} + O(ve^{-v}) + O(v^3e^{-v}).$$

(d) By implicit differentiation on (11), we have

$$\begin{aligned}
\frac{d^2v}{du^2} \;&=\; \frac{2\,c(v)^2\,(e^v - 1) + 4u\,c'(v)\,c(v)\,v'\,(e^v - 1) + 2u\,c(v)^2\,e^v v'}{2v\,c(v)\,(e^v - 1) - v^3} \\
&\qquad - v' \cdot \frac{2v'\,c(v)\,(e^v - 1) + 2v\,c'(v)\,v'\,(e^v - 1) + 2v\,c(v)\,e^v\,v' - 3v^2v'}{2v\,c(v)\,(e^v - 1) - v^3} \\
&=\; \left(\frac{c(v)}{v} + O(ve^{-v}) + \frac{uc(v)v'}{v}\right)(1 + O(v^3e^{-v})) \\
&\qquad - v'\left(\frac{v'}{v} + O(ve^{-v}) + v' + O(ve^{-v})\right)(1 + O(v^3e^{-v})) \\
&=\; \frac{c(v) - (v')^2}{v} + v'\left(\frac{uc(v)}{v} - v'\right) + O(v^3e^{-v}) \;=\; O(v^3e^{-v}),
\end{aligned}$$

where we use $c'(v) = O(ve^{-v})$ and the various prior bounds. □

We will also need to control the various derivatives of v with respect to u. In particular, apriori we know v is smooth on $(0,\infty)$, *not* on $[0,\infty)$. The following lemma shows that we can extend the smoothness to $[0,\infty)$.

Lemma 28 (Behavior at 0)**.** *We have $c(v) = v + O(v^2)$ as $v \to 0$. Additionally, the function v is infinitely differentiable at 0 with Taylor expansion of*

$$v = u^2 + O(u^4).$$

Proof. A straightforward series expansion gives that as $x \to 0$,

$$\frac{x}{e^x - 1} \;=\; \frac{x}{x + \frac{x^2}{2} + O(x^3)} \;=\; 1 + O(x),$$

so integrating this from 0 to v gives

$$c(v) \;=\; \int_0^v \frac{x}{e^x - 1}\, dx \;=\; v + O(v^2).$$

By taking $v \to 0$ in (11), we conclude

$$2u\frac{du}{dv} \;=\; \frac{2v}{v + O(v^2)} - \left(\frac{v}{v + O(v^2)}\right)^2 \cdot (1 + O(v)) \;=\; 1 + O(v).$$

Therefore, by implicit differentiation, we have that as $u \to 0$,

$$\frac{dv}{du} \;=\; \frac{2u}{\frac{2v}{c(v)} - \frac{v^2}{c(v)^2}\frac{v}{e^v - 1}} \;=\; \frac{2u}{1 + O(v)} \;=\; 2u + O(uv).$$

This shows that v is continuously differentiable at 0 with $v'(0) = 0$. To show that v is infinitely differentiable at 0, we note that the derivative $\frac{dv}{du}$ can be written as a smooth function of v and u (see (11)). Thus, taking the derivative of this equation give that v is twice differentiable at 0. Repeating this argument gives that v is infinitely differentiable at 0, as desired.

Finally, since we have that $v = O(u^2)$, we get that $\frac{dv}{du} = 2u + O(u^3)$ and therefore $v = u^2 + O(u^4)$. □

A.3. **Proof of Proposition 9 (a).** We claim that f is differentiable and has bounded first derivative, which suffices. By Lemma 26, we get that f is differentiable on compact subsets of $(0,\infty)$ (the obstruction preventing us from immediately concluding smoothness at 0 is u in the denominator). Thus, it suffices to show that

- f is differentiable near 0.
- the derivative f' is bounded near ∞.

To do this, we first compute the logarithmic derivative of f, getting that

$$\frac{f'(u)}{f(u)} \;=\; -\frac{1}{u} + \frac{v'}{v} - \frac{e^{-v}\left(v' - u + \frac{1}{2}u^2 v'\right)}{2\left(1 - e^{-v} - \frac{1}{2}u^2 e^{-v}\right)}, \tag{12}$$

where $v' = \frac{dv}{du}$.

Differentiability near 0. First, we note that as $u \to 0$, we have that $v = u^2 + O(u^4)$, so we have that

$$\begin{aligned} f(u) &= \frac{1}{2^{3/2}\pi}(u + O(u^3)) \cdot \left(1 - (1 - u^2 + O(u^4)) - \tfrac{1}{2}(u^2 + O(u^4))\right)^{-1/2} \\ &= \frac{1}{2^{3/2}\pi}(u + O(u^3)) \cdot \frac{2^{1/2}}{u}(1 + O(u^2))^{-1/2} \;=\; \frac{1}{2\pi} + O(u^2). \end{aligned}$$

Plugging this into (12) and using Lemma 28 (which gives $v' = 2u+O(u^3)$ and $v = u^2+O(u^4)$), we get that

$$\begin{aligned} f'(u) &= (2\pi + O(u^2))\left(-\frac{1}{u} + \frac{2u+O(u^3)}{u^2} - \frac{e^{-v}(2u+O(u^3)-u+O(u^3))}{2(v-\frac{1}{2}u^2+O(u^4))}\right) \\ &= (2\pi + O(u^2))\left(-\frac{1}{u} + \frac{2u+O(u^3)}{u^2} - \frac{u+O(u^3)}{u^2+O(u^4)}\right) \\ &= (2\pi + O(u^2))\left(-\frac{1}{u} + \frac{2}{u} - \frac{1}{u} + O(u^2)\right) \;=\; O(u^2), \end{aligned}$$

so in fact, $\lim_{u\to 0} f'(u) = 0$.

Smoothness near ∞. As $u \to \infty$, we have

$$\begin{aligned} f(u) &= \frac{v}{2^{3/2}\pi u}\left(1 - e^{-v} - \tfrac{1}{2}u^2 e^{-v}\right)^{-1/2} \\ &= \frac{\pi}{2^{3/2}\pi\cdot\sqrt{6}}(1+O(ue^{-u})) \;=\; \frac{1}{4\sqrt{3}} + O(ue^{-u}). \end{aligned}$$

We look at the derivative formula from (12). As $v \to \infty$, by Lemma 27 we have that $v' = \frac{dv}{du}$ is bounded, so the right hand side converges to 0. Since we know that f is bounded from below, we get that $\lim_{u\to\infty} f'(u) = 0$, showing that f' is bounded in an interval $[M, \infty)$ for some large M.

A.4. **Proof of Proposition 9 (b).** From Lemma 27 (b), we get by a simple substitution that $\lim_{u\to\infty} g(u) = \pi\sqrt{2/3}$.

It remains to show that g is strictly increasing. Let $u' < u$. We first show that g is weakly increasing. Suppose $g(u) < g(u')$. Then by Theorem 7 (where we take the error term to be size 2 and $\frac{1}{2}$), for n sufficiently large we conclude

$$\begin{aligned} p(n, u\sqrt{n}) &< 2\cdot\frac{f(u)}{n}\cdot\exp(g(u)\sqrt{n}) \\ &< \frac{1}{2}\cdot\frac{f(u')}{n}\cdot\exp(g(u')\sqrt{n}) \;<\; p(n, u'\sqrt{n}) \;\le\; p(n, u\sqrt{n}), \end{aligned}$$

which is a contradiction.

Now to get strictly increasing, suppose $g(u) = g(u')$. Observe that $p(n,k) \;=\; p(n,k-1) + p(n-k,k)$ and thus by induction,

$$p(n,k) \;\ge\; \ell p(n-k, k-\ell). \tag{13}$$

Put $u_0 = \frac{u+u'}{2} \in (u', u)$ and let ℓ be a positive integer. Pick n such that $u^* = \frac{u_0\sqrt{n}-\ell}{\sqrt{n-u_0\sqrt{n}}}$ satisfies $u^* \in (u', u)$ which holds for all sufficiently large n since

$$\frac{u_0\sqrt{n}-\ell}{\sqrt{n-u_0\sqrt{n}}} \;=\; u_0 + O(n^{-1/2})$$

where the implicit constant can depend on both u_0 and ℓ. Then for sufficiently large n by Theorem 7,

$$\begin{aligned} 2\frac{f(u_0)}{n}\exp(g(u_0)\sqrt{n}) &> p(n, u_0\sqrt{n}) \\ &\ge \ell\cdot p(n-u_0\sqrt{n}, u_0\sqrt{n}-\ell) \qquad \text{(by (13))} \end{aligned}$$

$$\begin{aligned} &> \frac{\ell}{2} \cdot \frac{f(u^*)}{n} \cdot \exp\left(g(u^*) \cdot \sqrt{n - u_0\sqrt{n}}\right) \\ &\geq \frac{\ell}{2} \cdot \frac{f(u^*)}{n} \cdot \exp\left(g(u^*) \cdot (\sqrt{n} - u_0)\right). \end{aligned}$$

Now, since g is weakly increasing with limit $\pi\sqrt{2/3}$, we conclude g will be constant on $[u', u]$ taking on some value $\leq \pi\sqrt{2/3}$. Thus

$$f(u_0) \;\geq\; \frac{\ell}{4} \cdot \exp\left(-\frac{u_0\pi\sqrt{6}}{3}\right) \cdot \inf_{u^* \in [u', u]} f(u^*)$$

so picking some ℓ where this is false will give us a contradiction.

A.5. **Proof of Proposition 9 (c).** We break the proof into two parts.

The functions h' and h'' are bounded. First, observe that h is smooth and thus its derivatives are continuous and bounded on compact subsets of $(0, \infty)$. Write $v' = \frac{dv}{du}$ and $v'' = \frac{d^2v}{du^2}$. Then

$$g'(u) \;=\; -\frac{2v}{u^2} + \frac{2v'}{u} - \log(1 - e^{-v}) - \frac{uv'\,e^{-v}}{1 - e^{-v}}, \tag{14}$$

$$g''(u) \;=\; \frac{4v}{u^3} - \frac{4v'}{u^2} + \frac{2v''}{u} - \frac{2v'e^{-v}}{1 - e^{-v}} + \frac{u(v')^2e^{-v}}{(1 - e^{-v})^2} - \frac{u(v'')e^{-v}}{1 - e^{-v}}, \tag{15}$$

and the derivatives of h are given by $h'(t) = g(1/t) - \frac{g'(1/t)}{t}$, and $h''(t) = \frac{g''(1/t)}{t^3}$.

Near $t = 0$, observe that $g(1/t)$ is continuous and bounded, and by Lemma 27, we have $\frac{v}{u} = \frac{\pi}{\sqrt{6}} + O(t^{-1}e^{-1/t})$, $v' = \frac{\pi}{\sqrt{6}} + O(t^{-3}e^{-1/t})$, and $v'' = O(t^{-3}e^{-1/t})$. Inserting all of these into (14) and (15) gives

$$\begin{aligned} \frac{g'(1/t)}{t} &= -2\left(\frac{\pi}{\sqrt{6}} + O(t^{-4}e^{-1/t})\right) + 2\left(\frac{\pi}{\sqrt{6}} + O(t^{-4}e^{-1/t})\right) + O\left(t^{-1}e^{-1/t}\right) + O\left(e^{-1/t}\right) \\ &= O(t^{-4}e^{-1/t}) \end{aligned}$$

and

$$\begin{aligned} \frac{g''(1/t)}{t^3} &= \frac{4}{t}\left(\frac{\pi}{\sqrt{6}} + O(t^{-3}e^{-1/t})\right) - \frac{4}{t}\left(\frac{\pi}{\sqrt{6}} + O(t^{-3}e^{-1/t})\right) + O(t^{-5}e^{-1/t}) \\ &\quad + O(t^{-3}e^{-1/t}) + O(t^{-4}e^{-1/t}) + O(t^{-7}e^{-2/t}) \\ &= O(t^{-5}e^{-1/t}), \end{aligned}$$

and thus h' and h'' are bounded in some interval of the form $(0, M]$ for $M > 0$.

Thus, it remains to consider near $t = \infty$ (so $u \to 0$). To do this, we look at (14) and note that as $u \to 0$, we have that every term in $\frac{g'(1/t)}{t} = ug'(u)$ is bounded. (In particular, $\frac{v}{u}$, v', $u\log(1 - e^{-v}) \sim u \log v$, and $\frac{u^2v'e^{-v}}{1-e^{-v}} \sim \frac{u^2v'}{v}$ are all bounded as $u \to 0$). This implies that $h'(t)$ is bounded as $t \to \infty$.

Similarly, every term in $\frac{g''(1/t)}{t^3} = u^3g''(u)$ is also bounded. The terms worth paying attention are the terms with $1 - e^{-v} \sim v$ in the denominator. These terms are

$$\frac{2u^3v'e^{-v}}{1 - e^{-v}} \sim \frac{2u^3}{v}, \quad \frac{u^4(v')^2e^{-v}}{(1 - e^{-v})^2} \sim \frac{u^4}{v^2}, \quad \text{and} \quad \frac{u^4v''e^{-v}}{1 - e^{-v}} \sim \frac{u^4}{v},$$

and they are all clearly bounded since $v \sim u^2$. Thus, we conclude that $h''(t)$ is bounded as $t \to \infty$, completing the proof.

The function h is strictly increasing. Observe that

$$\sqrt{n} \cdot g\left(\frac{k}{\sqrt{n}}\right) \;=\; k \cdot h\left(\frac{\sqrt{n}}{k}\right)$$

and so

$$\begin{aligned} p(n,k) \;&\sim\; \frac{f\left(\frac{k}{\sqrt{n}}\right)}{n} \exp\left(k \cdot h\left(\frac{\sqrt{n}}{k}\right)\right) \\ p\left(\frac{k^2}{t^2}, k\right) \;&\sim\; \frac{f(t)}{n} \exp\left(k \cdot h\left(\frac{1}{t}\right)\right). \end{aligned}$$

Let $t' < t$ and let ℓ be a positive integer. Take k sufficiently large so that $\frac{k^2}{(t')^2} - k > \frac{k^2}{t^2}$. Put $t^* = \frac{t(k-\ell)}{k}$. By Theorem 7, we have

$$\begin{aligned} 2 \cdot \frac{f(t')(t')^2}{k^2} \cdot \exp\left(k \cdot h\left(\frac{1}{t'}\right)\right) \;&>\; p\left(\frac{k^2}{(t')^2}, k\right) \\ &\geq\; \ell p\left(\frac{k^2}{(t')^2} - k, k - \ell\right) \qquad \text{(using (13))} \\ &\geq\; \ell p\left(\frac{k^2}{t^2}, k - \ell\right) \\ &\geq\; \frac{\ell}{2} \cdot \frac{f(t^*)t^2}{k^2} \cdot \exp\left(k \cdot h\left(\frac{1}{t^*}\right)\right). \end{aligned}$$

If $h\left(\frac{1}{t}\right) > h\left(\frac{1}{t'}\right)$ then by continuity of h for k sufficiently large we have $h\left(\frac{1}{t^*}\right) > h\left(\frac{1}{t'}\right)$. Thus, for k sufficiently large we have

$$2 \cdot f(t')(t')^2 \;\geq\; \frac{\ell}{2} f(t^*)t^2 \;\geq\; \frac{\ell}{4} f(t)t^2$$

where the second inequality is from the continuity of f and observing that $t^* \to t$ as $k \to \infty$. Picking some value of ℓ where this is false will give us a contradiction. Thus, we conclude that h is weakly increasing.

If $h\left(\frac{1}{t}\right) = h\left(\frac{1}{t'}\right)$ then by weakly increasing, h is constant on $[\frac{1}{t}, \frac{1}{t'}]$. Now if we repeat the above manipulation with t' and $\frac{t'+t}{2}$ then for k sufficiently large we have $t^* \in [t', t]$ and thus $h\left(\frac{1}{t'}\right) = h\left(\frac{1}{t^*}\right)$. Simplifying, we conclude

$$f(t')(t')^2 \;\geq\; \frac{\ell}{4}\left(\frac{t'+t}{2}\right)^2 \inf_{t^* \in [\frac{1}{t}, \frac{1}{t'}]} f(t^*),$$

and picking some value of ℓ where this is false will give us a contradiction.

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